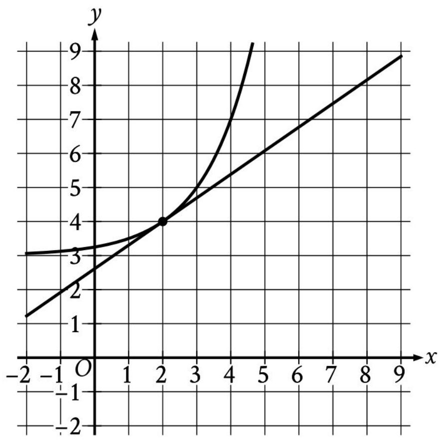
SAT Math

Nonlinear Equations and Systems 1

Question #	ID		Answer
1.1	3c95093c	$6x - 9y > 12$ Which of the following inequalities is equivalent to the inequality above? A. $x - y > 2$ B. $2x - 3y > 4$ C. $3x - 2y > 4$ D. $3y - 2x > 2$	B
1.2	4ca30186	 The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system? A. $(0, 0)$ B. $(0, 2)$ C. $(2, 4)$ D. $(4, 0)$	C
1.3	3de7a7d7	Which of the following is a solution to the equation $2x^2 - 4 = x^2$? A. 1 B. 2 C. 3 D. 4	B

Question #	ID		Answer
1.4	70f98ab4		B

$$q - 29r = s$$

The given equation relates the positive numbers q, r, and s. Which equation correctly expresses q in terms of r and s?

A. $q = s - 29r$

B. $q = s + 29r$

C. $q = 29rs$

D. $q = -\frac{s}{29r}$

1.5	568aaf27		D

$x + y = 12$
$y = x^2$

If (x, y) is a solution to the system of equations above, which of the following is a possible value of x?

A. 0

B. 1

C. 2

D. 3

1.6	b76a2815		A

$$P = \frac{W}{t}$$

The power P produced by a machine is represented by the equation above, where W is the work performed during an amount of time t. Which of the following correctly expresses W in terms of P and t?

A. $W = Pt$

B. $W = \frac{P}{t}$

C. $W = \frac{t}{P}$

D. $W = P + t$

1.7	c7789423		11, -7

$$|x - 2| = 9$$

What is one possible solution to the given equation?

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Question #	ID		Answer
1.8	eb268057	$x^2 = 64$ Which of the following values of x satisfies the given equation? A. -8 B. 4 C. 32 D. 128	A
1.9	98f735f2	The total revenue from sales of a product can be calculated using the formula $T = PQ$, where T is the total revenue, P is the price of the product, and Q is the quantity of the product sold. Which of the following equations gives the quantity of product sold in terms of P and T? A. $Q = \frac{P}{T}$ B. $Q = \frac{T}{P}$ C. $Q = PT$ D. $Q = T - P$	B
1.10	fcb78856	$$b = 42cf$$ The given equation relates the positive numbers b, c, and f. Which equation correctly expresses c in terms of b and f? A. $c = \frac{b}{42f}$ B. $c = \frac{b-42}{f}$ C. $c = 42bf$ D. $c = 42 - b - f$	A

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Question #	ID		Answer
1.11	4236c5a3	If $(x+5)^2 = 4$, which of the following is a possible value of x ? A. 1 B. -1 C. -2 D. -3	D
1.12	f11ffa93	$\sqrt{x+4} = 11$ What value of x satisfies the equation above?	117
1.13	5639dd1a	$x^2 = (22)(22)$ What is the positive solution to the given equation?	22
1.14	c1964c11	$p + 34 = q + r$ The given equation relates the variables p, q, and r. Which equation correctly expresses p in terms of q and r? A. $p = q + r + 34$ B. $p = q + r - 34$ C. $p = -q - r + 34$ D. $p = -q - r - 34$	B
1.15	7cb3a8ee	$\lvert x - 5 \rvert = 10$ What is one possible solution to the given equation?	15, -5

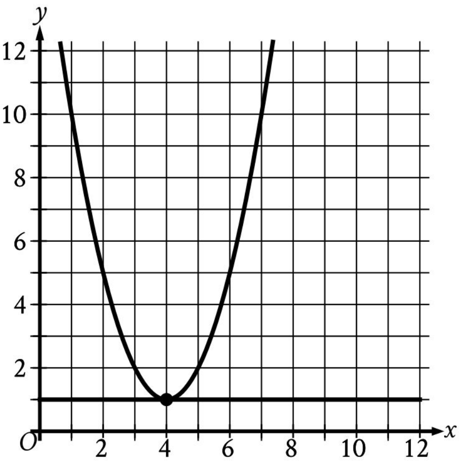
Question #	ID		Answer
1.16	b8c4a1cd	$$8j = k + 15m$$ The given equation relates the distinct positive numbers j, k, and m. Which equation correctly expresses j in terms of k and m? A. $j = \frac{k}{8} + 15m$ B. $j = k + \frac{15m}{8}$ C. $j = 8(k + 15m)$ D. $j = \frac{k+15m}{8}$	D
1.17	7a8cb72a	$7m = 2(n + p)$ The given equation relates the positive numbers m, n, and p. Which equation correctly gives m in terms of n and p? A. $m = \frac{2(n+p)}{7}$ B. $m = 2(n + p)$ C. $m = 2(n + p) - 7$ D. $m = 2 - n - p - 7$	A

Question #	ID	Answer
1.18	d0e8e8f5	D

Question ID d0e8e8f5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■□□

ID: d0e8e8f5



The graph of a system of a linear and a quadratic equation is shown. What is the solution (x, y) to this system?

A. $(0, 0)$

B. $(-4, 1)$

C. $(4, -1)$

D. $(4, 1)$

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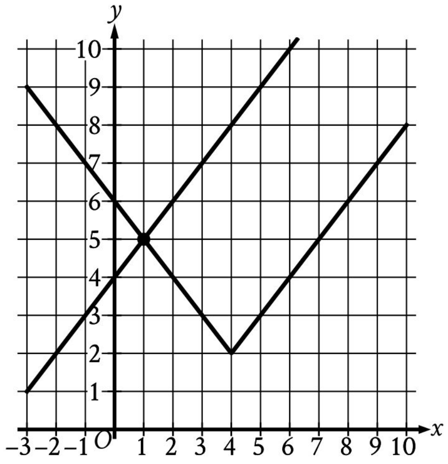
Question #	ID		Answer
1.19	d3f7c429		C



The graph of a system of an absolute value function and a linear function is shown. What is the solution (x, y) to this system of two equations?

A. $(-1, 5)$

B. $(0, 4)$

C. $(1, 5)$

D. $(4, 2)$

1.20	13e5a5d5		9

$$5|x| = 45$$

What is the positive solution to the given equation?

1.21	58443765		B

$$y = 5x + 4$$
$$y = 5x^2 + 4$$

Which ordered pair (x, y) is a solution to the given system of equations?

A. $(0, 0)$

B. $(0, 4)$

C. $(8, 44)$

D. $(8, 84)$

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Question #	ID		Answer
1.22	332cd67b	$$3x^2 - 15x + 18 = 0$$ How many distinct real solutions are there to the given equation? A. Exactly one B. Exactly two C. Infinitely many D. Zero	B